

Cohort 5, 2024-2026

Master's Thesis Guidelines

A. The master's thesis rationale

In doing the master's project, the student must utilise and demonstrate the knowledge and maturity acquired during the master's programme. The purpose of the thesis is to test the student's ability to combine their knowledge and skills in solving challenging problems in a scientific way.

In addition, the work on the master's project and thesis will develop student's ability to:

- Independently and under realistic circumstances, solve a concrete, well-defined problem within the area of the subject (e.g., spectral or colour imaging);
- Document this work in a report that describes the student's achievements and that demonstrates his/her abilities to present the project in a thorough and easy to comprehend manner, and
- Orally present his/her work.

The master's thesis gives **30 ECTS** credits, which correspond to six months of full-time work.

A.1 Scientific work

The following demands must be fulfilled in a scientific report:

- Statements that are not evident should be substantiated. This may be done by logical reasoning or by referring to a source.
- The work and the achievements must be compared and related to relevant work reported in the literature and to results that have been published in the research literature.
- Facts must clearly distinguish from interpretations of facts.
- Reported results must clearly be distinguished from results that other persons have achieved.
- Reported work must clearly distinguish from what has been done by other people.
- The source of facts and information found elsewhere must be referenced correctly and completely. This is also true when summarising someone's work or ideas.
- The report must describe not only the achieved results but also how results were achieved. The description of the methods must be clearly separated from the

description of the results.

- Problems that are encountered must be described honestly; the report must document how the problems were overcome, were worked around, or were left unsolved because no solution was found or because time ran out before a solution was found.

A.2 The report

- The report should be written for fellow students as well as for people working in the field. It should contain a description of the problem and the problem area as well as the methods used and the theory behind them. Large blocks of theory may be presented very briefly if a reference to a textbook is included.
- The report is public and COSI coordinating institution will publish the report on the web (unless stated otherwise in the Master's Thesis Agreement). Only parts of the report shall be kept confidential.
- The report must clearly state that it is a master's thesis defended at the Norwegian University of Science and Technology (NTNU) in Gjøvik and where the work has been performed and who has been the COSI and external supervisor(s).
- The title of the report is important. It should give a good description of the work, be understandable, and preferably sound interesting. Unusual abbreviations must be avoided and unusual terminology must be explained

A.2.1 The following formal requirements are put on the report:

- The report should be written in correct English, and it should have a title and an abstract. It may have a title and abstract in other languages as well.
- There is no fixed number of pages for the report. The reports should have between 50 and 70 pages plus appendixes, but the length of the report depends not only on the type of work that has been done but also on the writing style of the author. The report should be written using word or LaTeX which the LaTeX template will be provided to the student by the program. The report should have a suitable typographical layout and be delivered in PDF on the Inpera platform at NTNU.
- The reports will be published on the web (parts of the report can remain confidential if stated in the Master's Thesis Agreement or specific confidentiality agreement). This means that they will be printed either one-sided or double-sided depending on the printer used. The layout must work in both cases. Thus, do not number the introductory pages (abstract etc.) and let the first page of the first chapter be page number one.
- The report should introduce the problem area as well as an introduction to relevant theory. It should clearly state the problem being studied, describe the methods used to solve it and the criteria used to choose those methods, describe the results achieved, present the student's conclusions based on these results, and what likely effects the study may cause. The order may be different than the above.
- A detailed thesis structure and typography are annexed at the end of this document. The report should be carefully proofread. The number of typing mistakes etc. should

be very low. Hyphenations must be correct.

- The report should have a table of contents and the pages must be numbered.
- Illustrations and diagrams should have a legend under the illustration/diagram. Give each illustration a number.
- Tables should have table headers and should be numbered.
- Illustrations, diagrams, and tables should be placed close to the corresponding text. The text should refer to the illustration so that the reader knows when to look at it and what it illustrates (e.g. see figure 1). All illustrations must be of good typographic quality suited for presentation on paper. The author should bear in mind that printouts usually are made in grayscale.
- When presenting facts read elsewhere the author must give a reference. A reference is needed also if one gives a summary of someone's work.
- There must be a consistent and complete literature list.
- If program listings are included, they should be put in the appendix.

Material that gives additional details to the work but is not needed to understand the work may be put in appendices (questionnaires, instructions that the student has written etc.). Appendices must be numbered (Appendix 1, Appendix 2 etc.). All pages must also be numbered. Each appendix may start from page 1, or there may be a consecutive numbering through the report and all appendices.

A.2.3 Template:

An example of pre-defined template for use in LaTeX can be downloaded from the Blackboard platform. Students are asked to not change anything in the style of the template so that all thesis are consistent in style.

B. Organisation and evaluation of the master's thesis

B.1 The master's project

A master's project is the work of an individual student. Different students may solve problems that are closely related, but they must write individual reports. The master's project consists of two parts: the investigation the students make and the report they write. Both parts are important. The following requirements relate to the investigation part, and must be fulfilled during the project:

- At the beginning of the project, the student makes a specification of the project including a schedule and a reading list.
- In liaison with Katrine Moe (who is in charge of the COSI Master's Thesis Agreements) the student organises to complete thoroughly the Master's Thesis Agreement and to **obtain all required signatures by the 30th of November 2025.**

- Prior to the oral presentation at the thesis defence seminar, the draft of the master's thesis must be approved by the academic supervisor.
- The student must give an acceptable oral presentation of the master's thesis at the concluding master's defence seminar.

B.2 Role of the host supervisor

- Each student has a supervision team that consists of the **internal COSI supervisor(s)** and in the case of projects performed in organisations other than the four full partners an **external supervisor(s)**. The supervision team judges whether the proposed project is suitable for a master's thesis. The student must have regular contact with his/her supervision team. When the master's project is based in an external institution or company, it is recommended that the internal COSI supervisor visits both the student and the external supervisor in the workplace at least once.
- **The student is responsible for keeping regular and frequent contact with his/her supervision team** and giving them reasonable time to read the report. As a general rule, it is reasonable to count at least 10 days for a thorough read.
- The internal COSI supervisor is responsible for evaluating the progress of the student. Failing to receive approval of progress for two (2) consecutive months or three (3) non-consecutive months will result in the failure of the master's thesis
- The draft of the master's thesis should be formally approved by the academic supervisor at least two (2) weeks prior to the submission date of the master's thesis. Failure to do so will result in the failure of the master's thesis. The draft approved by the academic supervisor can be modified by the student after approval.
- The **supervision team** will evaluate the student for his/her research skills, personal and communication skills, ability to work in a team and progress made along the master's thesis. The provided feedback would then be used when in the AMB evaluation of the master's thesis. See template "*COSI_Supervision_Team_Evaluation*".

B.3 The master's thesis

Throughout the fourth semester, students are expected to work in close consultation with an internal COSI academic supervisor(s), forming a coherently argued thesis that demonstrates scholarly research skills and knowledge of the selected area of study. It is recommended to discuss the structure of the thesis with the supervisor well before starting on the report writing.

The Academic Management Board (AMB) is responsible for evaluating the progress of the student during his/her master's thesis. **40% of the overall grade will be allocated to this issue.** See template "*COSI_AMB_Evaluation*".

B.3.1 The master's thesis public presentation

- All master's projects must be presented orally at the master's thesis defence seminar. The presentations are public and held in Gjøvik.
- The **30 minutes** presentation of the draft Thesis should include an introduction to the problem, a description of the actual project, the results achieved, the student's interpretation of the results, and a discussion of what influence the results may have. Questions from the reviewers as well as the audience will follow the presentation.
- The master's thesis presentation slides must be uploaded on Blackboard at the latest by **the 27th of August 2026**.
- The thesis presentations will take place on the **31st of August and 1st of September 2026**.
- The master's thesis presentation will be scored by two reviewers who will complete an evaluation form which will be returned to the AMB at the latest the **next working day after the presentation**.
- **The master's thesis public presentation represents 20% of the overall fourth-semester grade.** See "*COSI_Master_Thesis_Oral_Presentation_Evaluation*".

B.3.2 The master's thesis Report

- Students are expected to follow the requirements and recommendations for corrections given by their supervision team. If any doubts as to what to change, the student should ask the internal COSI supervisor.
- At the latest, by the **15th of July 2026**, students must submit their report on Inspira. The final version must be approved by the Internal COSI. The submitted theses will be scored by two reviewers.
- The reviewers will complete an evaluation form which the feedback will be returned to students by the **15th of August 2026**, before the master's thesis public presentation. **The report will represent 40% of the overall fourth-semester grade.** See template "*COSI_Master_Thesis_Report_Evaluation*".

B.4 Failing the master's thesis

- A master's thesis will fail and be graded F (for further details about grading look to the master's thesis rules) if:
 - It is not submitted in time,
 - The student did not fulfil the requirements listed in section A. 2.1, or
 - The quality of the submitted thesis is below what is considered sufficient to pass.
- Any grade below C (D to E) on the master's thesis will entitle the student to only validate the national diploma issued by full partner universities according to the student's mobility track, but not the EMJMD diploma.
- A student who fails to pass the master's thesis has the right to a second attempt. In this case, the student can:
 - Find a new topic for the second attempt and must write an application,

including a new project plan, for this new project. The same or a different supervisor will be appointed, depending on the new topic.

- Use the same topic if this has been agreed in advance by the Joint Graduate committee members. The same or a different supervisor could be appointed depending on the situation.
- The second attempt (to obtain the 30 ECTS of the master's thesis) will take place the year following the failure and the student will have to register again in person at NTNU and pay any local fee at NTNU at the beginning of the year. The student could complete his/her master's thesis at a distance, keeping regular contact and visits with the supervision team and will need to present his work at NTNU during the master's thesis defence seminar which will be organised in August or September the following academic year.
- The student forfeits the right to complete the master's programme if failing to pass on the second attempt.

B.5 Appeal process

- Students have three weeks to appeal the master thesis grade from the time the grade was announced.
- Students will not receive specific grades for each part of the assessment and the final average letter grade is the only grade that can be provided.
- Students do not have a right to appeal parts of the assessment elements of the master thesis and if appealed the whole master thesis would need to be re-evaluated.
- Upon the appeal of the master thesis grade:
 - Two new reviewers will be assigned to evaluate the thesis report
 - The supervision team will be contacted again to provide a new evaluation form.
 - A QAB member will be assigned to the project to provide a review of the work and chair the new presentation session.
 - A new presentation session will be held in order for the new reviewers to evaluate the thesis presentation.
 - Although we would prefer to have the presentation done in person but digital presentations could also be possible in special cases.
- The new grade given after the appeal process will replace the initial grade given.
- In cases that there is a gap of two grades or more (higher or lower) between the grade after the appeal and the initial grade, a third assessment will be conducted by another reviewer.